

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech.

Branch: Civil Engineering

Semester: III

Subject Code & Name: BTCVC305 Surveying

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is NOT a part of a chain surveying instrument?	L2, CO1	1
	a. Ranging rod b. Prismatic compass c. Chain d. Peg		
2	Inaccurate ranging leads to:	L2, CO1	1
	a. Reduction in chaining time b. Increase in chaining accuracy c. Errors in measured distance d. None of the above		
3	In compass surveying, the angle measured clockwise from the north is called:	L2, CO1	1
	a. Reduced bearing b. Whole circle bearing c. Magnetic declination d. Azimuth		
4	The instrument used to sight objects in plane table surveying is:	L2, CO2	1
	a. Dumpy Level b. Alidade c. Theodolite d. Cross Staff		
5	The primary objective of leveling is to determine:	L2, CO2	1
	a. Horizontal distances b. Differences in elevation c. Angular measurements d. Contours of land		
6	Leveling is widely used in:	L2, CO2	1
	a. Highway construction b. Earthwork calculation c. Contour mapping d. All of the above		
7	The least count of a Vernier theodolite is typically:	L2, CO3	1
	a. 20 seconds b. 20 minutes c. 20 degrees d. 20 radians		
8	If the horizontal circle reading of the theodolite is 30° for one station and 90° for another, the angle between the two stations is:	L2, CO3	1
	a. 30° b. 60° c. 90° d. 120°		
9	Which of the following is NOT a type of engineering survey?	L2, CO3	1
	a. Route survey b. Hydrographic survey c. Geological survey d. Astronomical survey		
10	Aerial surveys are typically conducted using:	L2, CO3	1
	a. Drones, planes, and satellites b. Theodolites and leveling instruments c. Dumpy levels d. Handheld compasses		
11	For construction of dams, which type of survey is most important?	L2, CO3	1
	a. Geological survey b. Route survey c. Astronomical survey d. Astrologic survey		

12	Which of the following is a key application of engineering surveys?				L2, CO3	1
	a. Land subdivision	b. Urban infrastructure planning	c. Irrigation system design	d. All of the above		
Q. 2	Solve the following.					12
A)	Describe the equipment used in chain surveying.				L2, CO1	6
B)	Explain the procedure of chain surveying.				L2, CO1	6
Q. 3	Solve the following.					12
A)	What are the advantages and disadvantages of compass surveying?				L2, CO1	6
B)	What are the accessories used in plane table surveying, and their functions?				L2, CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain the temporary adjustments of a leveling instrument.				L2, CO2	6
B)	What is the difference between the height of instrument (HI) and rise and fall methods?				L4, CO2	6
C)	The following readings were taken during leveling using the Height of Instrument (HI) method. The instrument was shifted after the 3rd and 6th readings. The first reading was taken on a benchmark (BM) with an RL of 100.00 m. Calculate the Reduced Levels (RL) of all points and check for arithmetic correctness.				L3, CO2	6
	Stn	BS	IS	FS		
	1	1.500				
	2		1.200			
	3		0.800	0.500		
	4	1.400				
	5		1.000			
	6		0.600	0.800		
	7	1.600		0.500		
Q. 5	Solve Any Two of the following.					12
A)	A theodolite is used to measure the horizontal angle between two lines AB and BC. The observations are:				L3, CO3	6
	Angle between AB and BC = $75^{\circ} 30' 40''$, Bearing of line AB = $120^{\circ} 45' 00''$					
	Calculate the bearing of line BC.					
B)	Write short notes on: Traversing by theodolite, Angle measurement.				L3, CO3	6
C)	What are the sources of errors in theodolite surveying? How can they be minimized?				L3, CO3	6
Q. 6	Solve Any Two of the following.					12
A)	What is the importance of engineering surveys in construction projects?				L4, CO3	6
B)	How do engineering surveys contribute to urban planning?				L4, CO3	6
C)	Write a Note on Geodetic Surveying.				L3, CO3	6

*** End ***